

Zaighum Zawar

zaighumzawar7@gmail.com | linkedin.com/in/Zaighum-Zawar | github.com/Zaighum003

PROFESSIONAL SUMMARY

Junior Full Stack AI Engineer with 1+ year of startup experience designing and deploying AI-powered applications using LLMs, Generative AI, Agentic AI systems, and Retrieval-Augmented Generation (RAG) pipelines. Proficient in Python, LangChain, LangGraph, FastAPI, React, and RESTful APIs, with practical exposure to GCP and Azure. Strong foundation in context engineering, prompt engineering, vector databases, MLOps, and system design — with a track record of delivering scalable, production-ready solutions.

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn), JavaScript, SQL, PHP, HTML/CSS

AI / ML: LangChain, LangGraph, RAG, Vector Databases (pgvector), Prompt Engineering, Context Engineering, MLOps

Web & APIs: FastAPI, React, Node.js, RESTful APIs, WebSockets, Microservices, Docker

Databases & Cloud: PostgreSQL, MySQL, GCP, Azure, Git/GitHub

Tools: N8N, Claude Code, Cursor, Jupyter Notebook, Postman, Linux, GitHub Copilot, MCPs, VS Code

PROFESSIONAL EXPERIENCE

Co-Founder & AI Engineer

Jan 2024 – Present

Botanex Forge

Remote

- Led development of advanced automation workflows and Python automation bots that streamlined client business operations end-to-end.
- Co-developed and deployed full-stack web and mobile applications for clients using React, FastAPI, and PostgreSQL on cloud infrastructure.
- Designed and integrated Agentic AI systems and LLM-powered pipelines into client products leveraging LangChain, LangGraph, and MCP protocols.
- Collaborated on architecture decisions for scalable microservices and RESTful API design across multiple concurrent client projects.

PROJECTS

Agentic Clinical Decision Support System (CDSS)(FYP)

Python, LangChain, LangGraph, RAG, ChromaDB

- Developed an AI-powered Clinical Decision Support System (CDSS) leveraging agentic workflows, Retrieval-Augmented Generation (RAG), and multi-agent reasoning to deliver real-time, evidence-based clinical recommendations.
- Engineered custom PubMed and web-search MCP servers to ingest, preprocess, and index medical literature into ChromaDB, enabling accurate semantic retrieval across large-scale healthcare knowledge bases.
- Integrated an RxNorm MCP server for automated drug-drug interaction analysis and medication validation, improving the reliability and safety of generated clinical recommendations.
- Implemented context-aware vector retrieval pipelines using ChromaDB and LangChain to enhance grounding accuracy, reduce hallucinations, and improve response relevance in clinical workflows.

Blog Generation Agentic AI App

FastAPI, LangGraph, LangChain, Groq

- Engineered an Agentic AI blog generator using FastAPI and LangGraph with graph-based routing for topic-driven automated content generation.
- Implemented multilingual translation paths (French and Urdu) via LLM chaining, expanding content accessibility across diverse audiences.

AI-Powered Vehicle Predictive Health Maintenance

Python, WebSockets, PostgreSQL, pgvector, RAG

- Built a real-time vehicle diagnostics platform ingesting live OBD-II data via WebSockets into an agentic AI reasoning core.
- Implemented RAG with pgvector for car-specific fault analysis and repair recommendations, improving diagnostic accuracy and speed.

EDUCATION

National University of Computer and Emerging Sciences (NUCES FAST)

Aug 2022 – Present

Bachelor of Science in Computer Science (BsCS)

GPA: 2.96 / 4.0

The City School, Gulshan-e-Iqbal Campus, Karachi

Jun 2020 – Jun 2022

A Levels (Advanced Levels)

The City School, Johar Campus, Karachi

Jun 2017 – Jun 2020

O Levels (Ordinary Levels)

PROFESSIONAL CERTIFICATIONS

GCP Generative AI Leader – Google Cloud Platform

Agentic AI with LangChain and LangGraph – IBM

Building AI Agents and Agentic Workflows Specialization – IBM

Machine Learning Specialization – Coursera / DeepLearning.AI

GitHub Foundations Certification – GitHub

Google Prompting Essentials – Google

Web Designing for Everyone – University of Michigan